$$\begin{aligned}
\psi(u,u) &= 0 \\
\psi(u,d) &= \frac{1}{\sqrt{2}} \\
\psi(d,u) &= \frac{1}{\sqrt{2}} \\
\psi(d,d) &= 0.
\end{aligned}$$

Next, we'll use these four equations to calculate each element of Alice's density matrix by expanding the summation of Eq. 7.23. In the expansion, notice that for every factor of the form $\psi^*(a, b)\psi(a', b)$, Bob's input is the same for both factors. We discard any terms that do not have this property. This is what we mean by "setting b' equal to b in the summation." Here is the expansion:

$$\begin{aligned}
\rho_{uu} &= \psi^*(u,u)\psi(u,u) + \psi^*(u,d)\psi(u,d) = \frac{1}{2} \\
\rho_{ud} &= \psi^*(u,u)\psi(d,u) + \psi^*(u,d)\psi(d,d) = 0 \\
\rho_{du} &= \psi^*(d,u)\psi(u,u) + \psi^*(d,d)\psi(u,d) = 0 \\
\rho_{dd} &= \psi^*(d,u)\psi(d,u) + \psi^*(d,d)\psi(d,d) = \frac{1}{2}.
\end{aligned}$$

These values are the elements of a 2 × 2 matrix: